

Memory Tree Guided Key Frame Querying for Efficient 3D Question Answering

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Abstract. Answering questions accurately and efficiently in embodied scenarios presents significant challenges due to limited computational and memory resources for Vision Language Model (VLM) inference. Existing methods adopt visual search key frame retrieval method to select critical question-related key frames for VLM input. However, visual search methods are inefficient because they require visual search among thousands of video frames for each individual user query. In this work, we propose a memory tree guided key frame selection paradigm for efficient 3D question answering in embodied scenarios. Our method leverages a compact and reusable 3D scene representation, termed MemTree3D, which supports real-time online construction leveraging camera 6-DoF poses. MemTree3D captures multi-level 3D scene information, enabling a Large Language Model to efficiently query and retrieve question-relevant key frames through our scoring-based frame selection without reprocessing the entire video stream. On OpenEQA, our method improves the LLM-Match of GPT-4o by 17.4%, LLaVA-OneVision-7B by 5.8%, outperforms existing visual search methods. Our code is available at <https://github.com/hsiangwei0903/MemTree3D>.

Keywords: 3D Question Answering · Embodied Question Answering

1 Introduction

3D question answering [5, 42, 44] is a fundamental task in the field of scene understanding. It plays a crucial role in bridging vision-language model systems with real-world 3D environments, enabling a wide range of downstream applications such as embodied navigation [8, 44, 58], household robotics [48], augmented and virtual reality [16, 38], and assistive agents [15, 26, 47] that require accurate spatial reasoning and grounded semantic understanding of complex scenes. However, performing 3D question answering efficiently and accurately has been challenging for real-world embodied systems due to restricted computational and GPU memory resources. Current state-of-the-art Vision-Language Models

* Work done during internship at Amazon.

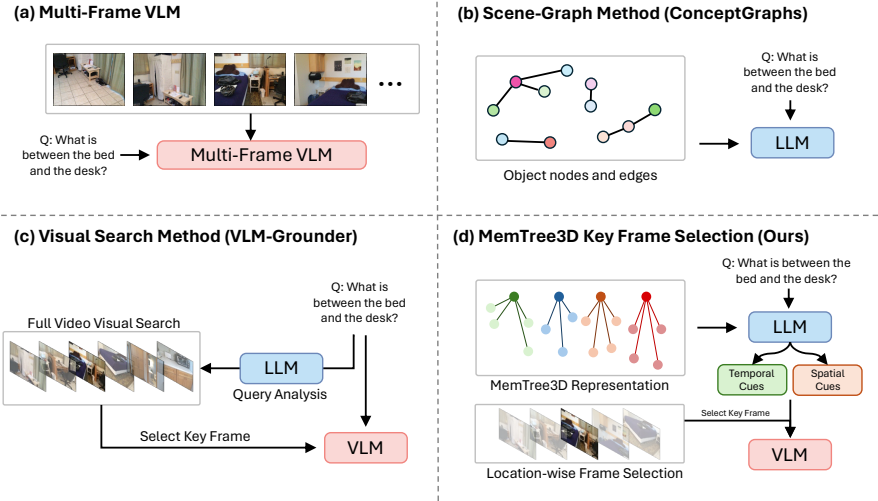


Fig. 1: Comparison of 3D question answering methods. (a) Multi-Frame VLM takes 3D scan video as input but suffers from proportional computation and memory cost scales with video length. (b) Scene-Graph Methods such as ConceptGraphs [18] leverages object-level scene graph and an LLM for question answering, but the coarse-level scene representation leads to visual information loss. (c) Visual Search Method such as VLM-Grounder [55] leverages object detector to perform visual search for each query, which leads to extensive runtime. (d) Our method adopts an LLM to reason over our designed MemTree3D representation and provide temporal and spatial cues for efficient, location-wise frame selection, bypassing the extensive visual search.

(VLMs) mostly adopt the typical transformer architecture [50], with the attention mechanism’s computation and memory requirements scaling quadratically with the input video length. This limitation creates a fundamental bottleneck, as existing 3D scene scan videos are typically captured at high FPS, which results in thousands of video frames and necessitates the use of frame sampling strategies for VLM-based 3D question answering.

Existing LLM/VLM-based 3D question answering methods can be broadly categorized into several mainstream approaches, including (a) Multi-Frame VLMs, (b) Scene-Graph Method, and (c) Visual Search Method, as shown in Figure 1. State-of-the-art multi-frame VLMs [33, 34] already possess the ability to perform 3D question answering, but are largely constrained by the computational and memory resources in the embodied scenario, often require video temporal downsampling strategies that leads to visual information loss. Scene-graph methods [7, 18, 44] leverage object-level scene representation for LLM question answering, the object-only representation, despite being simple and memory efficient, limits the system from performing fine-grained visual recognition beyond object-level understanding. To achieve a balance between memory constraint and more advanced visual recognition ability, visual search methods [55] adopt

an LLM to first perform query analysis, which identifies query-related objects and performs key object visual search in all video frames to select key frames for VLM input. This strategy reduces the memory overhead of VLM while preserving key frame information, yet still possess several major limitations, including 1) the visual search runtime scales with the video length, which hinders its ability for real-time user interaction 2) the visual search is required for each individual query, and cannot efficiently handle multi-round query scenarios, 3) the method does not account for visual search failure, where the vision model fails to retrieve meaningful query-related key frames from the video.

In this work, we identify and address two major limitations of existing visual search-based 3D question answering methods. First, these approaches rely heavily on the underlying vision model to perform visual search, yet lack a mechanism to recover from failures when query-relevant objects are missed. Second, they require exhaustive visual inference over all video frames, which must be repeated for every input query and significantly hinders real-time response. To alleviate these issues, we introduce an LLM-driven key-frame retrieval framework built upon a pre-constructed 3D scene representation. This design enables more flexible and robust query handling through the LLM’s explicit spatial and temporal reasoning capabilities. Furthermore, the system efficiently retrieves relevant key frames with respect to the user query without extensive visual search. Specifically, we leverage a tree-based architecture as a compact, reusable representation of the 3D scene, which can be efficiently queried by the LLM to generate informative cues for retrieving query-relevant key frames. These selected frames are then passed to a VLM to perform final question answering.

Our approach begins with the online and real-time construction of a hierarchical representation **MemTree3D**. MemTree3D encodes 3D scene at multiple levels, including frame-level detections, temporally-aware object relationships, and spatially localized segments derived from 6-DoF camera poses. This tree-based structure enables the LLM to reason over the scene content and identify critical frames without scanning every video frame. It is also reusable across multi-round queries: once constructed, the tree can be queried repeatedly by the LLM for different questions without re-running the vision model. Finally, it is robust to visual search failures. Because the LLM reasons over the symbolic and structural information in the tree (an example in Fig 2), it can still identify relevant frames even when query-related objects are missing. Extensive experiments on 3D question answering benchmarks show that our MemTree3D key frame selection paradigm improves state-of-the-art question answering accuracy, outperforms uniform sampling GPT-4o by 17.4%, largely reduces the runtime compared to visual search method by efficiency through MemTree3D querying and enhances robustness to perception failures via LLM-based reasoning.

In summary, the contribution of the work are three-folds:

1. We introduce **MemTree3D**, a compact and reusable 3D scene representation that supports real-time construction and enables LLM-driven key-frame selection, paving the way for scalable real-world embodied applications.

2. We propose a novel MemTree3D guided key frame selection paradigm for the 3D question answering task, significantly improving efficiency over existing visual search key-frame selection approaches.
3. We conduct extensive experiments and show that our method improves the LLM-Match of GPT-4o by 17.4% and achieves a 69.2% key-frame retrieval speedup compared to prior visual search methods, demonstrating a strong and efficient paradigm for 3D question answering.

2 Related Work

2.1 3D Question Answering

Capturing complex 3D scenes in a compact representation and conduct question answering remains a significant challenge in current research. Recent 3D-LLMs [56, 64, 65] leverage point cloud representations as LLM input for question answering task, but they require extensive 3D data training and still fall short in performance and generalizability compared to 2D models [25, 51, 61]. Object-centric representations [21, 23, 52] and 3D scene graphs [4, 7, 13, 18, 54] are also popular approaches, yet they suffer from perception failures such as missing objects or false positives. The coarse, object-level representation also limits scene-graph’s ability in answering questions related to 3D scene details. Furthermore, most existing scene graphs have inherently complex construction process, which require multiple foundation models [31, 41, 45] and often cannot support real-time construction. On the other hand, recent 2D video VLM [12, 36, 37, 60, 62, 63] directly utilize 3D scan videos as input for 3D scene question answering, the extensive training data enables 2D VLMs to achieve comparable performance with 3D-LLMs. But they face significant computational overhead due to the excessive number of frames in high FPS 3D scan videos. To address these challenges, we propose MemTree3D, a lightweight 3D scene representation that differs from traditional scene graphs by incorporating temporal information that supports key frame retrieval and real-time 25+ FPS construction. Rather than serving as the final scene representation, our proposed MemTree3D acts as an intermediate structure for LLM-driven key frame retrieval, mitigating the visual information loss in existing scene graph-based methods, and further address the challenges of computational overhead in video-LLMs by selecting question-related key frame for 3D question answering.

2.2 Key Frame Selection for Video Understanding

To address the computational and memory overhead in long-form video, recent research has developed key frame selection techniques to select question-relevant frames for VLM input. Most existing key frame selection methods follow the visual search key frame selection paradigm, which leverages the input question to identify relevant frames using techniques such as detecting question-related objects [55, 59], VLM-based selection [22, 58], or computing image-text similarity

scores [51] via vision foundation models [35, 45]. While these approaches generally outperform naive uniform sampling, the visual search paradigm significantly limits their efficiency in multi-round user query scenarios, as each query requires re-running visual search across the entire video. In this work, we investigate the setting of leveraging a one-time, pre-constructed high-level 3D scene representation for efficient, LLM-driven key frame retrieval. Our approach deviates from standard visual search key frame selection paradigm that suffers from the large computational cost scaling with the video length, our paradigm achieves highly efficient key frame selection that does not significantly degrade with the video length, while achieving SOTA 3D question answering performance.

3 Method

3.1 Overview

We propose the MemTree3D guided key-frame selection paradigm for the 3D question answering task, motivated by the need to bypass the extensive visual inference required by existing visual search-based methods, which must run a vision model over all video frames for every individual query. We illustrate the overview of our method in Fig 2. Given a 3D scan video, we first pre-construct a compact and lightweight 3D scene representation MemTree3D from the video. An LLM then takes the query together with MemTree3D as input to perform explicit reasoning and generate spatial and temporal cues for efficient key-frame retrieval. The selected key frames are subsequently passed to a VLM for final question answering. This paradigm offers two key advantages. First, it reduces overreliance on visual search results by enabling flexible, query-adaptive cue generation through LLM reasoning. Furthermore, it improves efficiency by eliminating repeated, query-dependent visual inference that scales with video length, replacing it with an LLM-based cue generation step with consistent runtime.

3.2 MemTree3D

We first introduce our proposed MemTree3D representation, a compact and lightweight 3D scene representation designed to support LLM reasoning and to provide spatial and temporal cues for key-frame selection. Despite the existence of several well-developed 3D scene representations, including point clouds [20, 56], feature fields [17, 21], and scene graphs [18, 29], each exhibits limitations that make it unsuitable for our method. Point cloud and feature field representations typically require additional LLM retraining, which incurs substantial computational and development cost. Existing scene graph-based methods such as ConceptGraphs [18], while capable of zero-shot LLM reasoning through textual serialization, rely on an extremely heavyweight construction pipeline that often combines multiple foundation models [31, 41, 45] and fail to achieve 10+ FPS real-time performance. Motivated by these limitations, we design **MemTree3D**, a **minimal** and **lightweight** 3D scene representation that

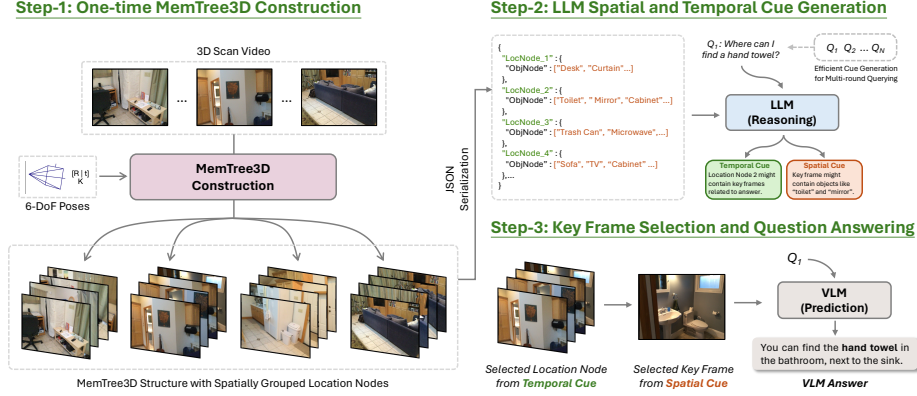


Fig. 2: Method Overview. Step-1: A compact MemTree3D representation is constructed once from the 3D scan video using 6-DoF camera poses to spatially group frames into location nodes. Step 2: The serialized MemTree3D structure is provided to an LLM, which performs reasoning to generate temporal and spatial cues for locating relevant scene regions. Step 3: Guided by these cues, informative key frames are retrieved and passed to a VLM to produce the final answer.

supports real-time (**25+ FPS**) construction while remaining **sufficient** for effective LLM reasoning and cue generation, which offers practicability for real-world deployment. MemTree3D is organized as a three-level tree structure. We describe the design and functionality of each level as follows:

Location Node. We split the 3D scene video into multiple segments and construct a location node (LocNode) for each segment, as shown in Fig 3. Each LocNode is assigned a unique location ID. While several existing works [19, 53] rely on image semantic to cluster video into segments, such approaches pose challenges in the embodied AI setting. These clustering methods are effective in general video question answering tasks, where inter-frame differences often correspond to changes in video content or camera shots. However, in the embodied scenario, the camera captures a continuous 3D scan with smaller semantic variation between adjacent frames, especially in visually uniform scene. This makes semantic-based clustering less reliable for identifying spatial transitions in 3D scenes. The second challenge lies in the computational overhead of extracting visual semantics, which relies on off-the-shelf vision foundation model. This further poses difficulties in resource-constrained embodied scenarios.

To address these limitations, we propose to segment the video stream in a way that (1) introduces minimal additional computational cost, and (2) ensures that each segment corresponds to a distinct location in the 3D scene. In our work, we utilize the 6-DoF pose, which is information that can be obtained directly from most existing embodied AI devices. We keep track of each frame’s 6-DoF pose during video processing and maintain a previous 6-DoF pose, P_{prev} , to calculate the translation and rotation with respect to the 6-DoF pose at the current

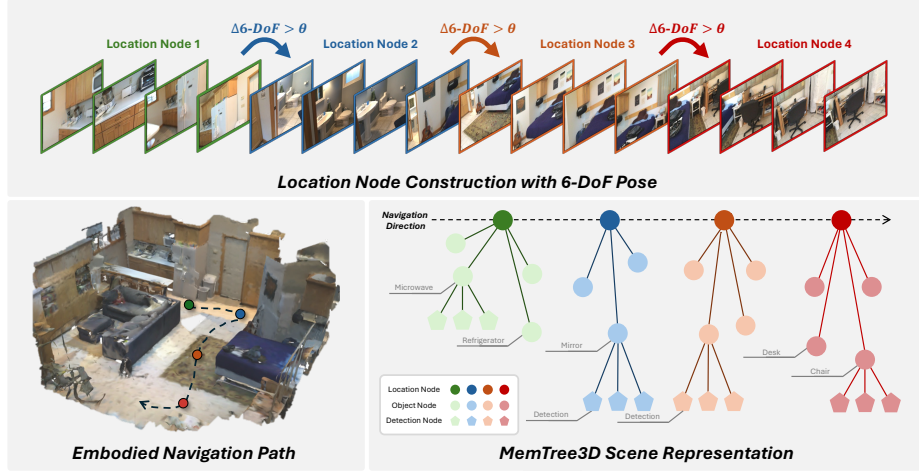


Fig. 3: Our proposed MemTree3D construction process. During the embodied navigation, the camera 6-DoF poses are used to construct the location node, each location node contains multi-level 3D scene information for later LLM key frame selection.

frame, P_t . Whenever the 6-DoF changes exceed translation threshold T_{thres} or rotation threshold R_{thres} , we construct a new LocNode and update P_{prev} with the current pose P_t ; we provide a detailed pseudo code in Algorithm. 1.

Object Node. In the 3D scene scan, we continuously leverage an object detector [11] and a multi-object tracker [2] to obtain object detections and their temporal dependencies across time stamps. In each location node, we can obtain multiple object tracklets from the detector and tracker. For each location, we wrapped each observed tracklet into an object node ObjNode. Each ObjNode stores a compact trajectory derived from the tracker. The combinations of ObjNode at each location ensure a compact, high-level coarse representation, which can enable the LLM querying to reason over the coarse perception content for each location in the scene.

Detection Node. Detection node (DetNode) is the leaf component in the MemTree3D structure, representing the detection in each frame. Each DetNode contains only the basic attribute, including the detection time-stamp, detection results like bounding box coordinates, and detection confidence representing the quality of the detection, which will be used in the later frame selection.

3.3 LLM Spatial and Temporal Cue Generation

After constructing MemTree3D, we obtain a compact 3D scene representation suitable for LLM reasoning. Given a user question Q , we leverage an LLM to reason over MemTree3D and generate temporal and spatial cues for subsequent

Algorithm 1: LocNode Construction

Input: Poses $P[0:N-1]$, Detections $D[0:N-1]$
Output: Location nodes $\mathcal{L}$

```

1  $\mathcal{L} \leftarrow \emptyset$ ,  $loc\_id \leftarrow 0$ ;
2  $P_{prev} \leftarrow P[0]$ ;
3  $d \leftarrow \emptyset$  // Detection buffer
4 for  $t \leftarrow 0$  to  $N-1$  do
5    $P_t \leftarrow P[t]$ ;
6    $d \leftarrow d \cup \{D[t]\}$ ;
7    $t_\Delta \leftarrow \Delta\text{Translation}(P_{prev}, P_t)$ ;
8    $r_\Delta \leftarrow \Delta\text{Rotation}(P_{prev}, P_t)$ ;
9   if  $t_\Delta > T_{thres}$  or  $r_\Delta > R_{thres}$  then
10    // Create new location node from buffered detections
11     $\mathcal{L} \leftarrow \mathcal{L} \cup \text{LocNode}(loc\_id, d)$ ;
12     $loc\_id \leftarrow loc\_id + 1$ ;
13     $d \leftarrow \emptyset$ ,  $P_{prev} \leftarrow P_t$ ;
13 return  $\mathcal{L}$ ;
```

key-frame selection. Specifically, we serialize the constructed MemTree3D representation into a textual JSON format and feed it to the LLM, as illustrated in Fig 2. This JSON representation includes only the first two levels of nodes, namely LocNode and ObjNode, while DetNode is omitted since fine-grained detection information is not essential for high-level cue generation. The LLM performs explicit reasoning based on the objects present in each location and to produce temporal and spatial cues that guide efficient frame retrieval.

Temporal Cues. The temporal cue generation focuses on identifying a small set of candidate locations (and their associated temporal segments) that are most likely to contain the answer to the query. When query-relevant objects are present, the LLM can directly associate the question with the corresponding locations by matching semantic object attributes. In more challenging cases where critical objects are missing due to perception failures, the LLM reasons over the relational and contextual information encoded in MemTree3D to infer the most plausible locations that may contain the answer. This reasoning-driven temporal cue generation enables robust recovery from perception errors, a capability that differentiates our work with existing visual search [55] or scene-graph-based approaches [18, 29], which typically fail when the underlying vision model cannot identify key objects.

Spatial Cues. In addition to temporal cues, the LLM generates spatial cues by predicting two sets of objects: key objects O_{key} and cue objects O_{cue} . Key objects correspond to entities that are directly relevant for locating the answer to the question, while cue objects denote surrounding or co-occurring objects that are likely to appear near the key objects in the scene. These spatial cues capture both direct and contextual object relationships and are subsequently used as

criteria in our scoring-based key-frame selection stage. By explicitly modeling spatial relationships through LLM reasoning, our approach reduces overreliance on exact object detections and provides flexible, query-adaptive guidance for efficient frame retrieval.

3.4 Key Frame Selection from Cues

Given the top- k temporal segments predicted by the LLM, we select one representative key frame from the video frames associated with each location, resulting in k query-relevant key frames. The goal is to identify frames that are maximally informative for answering the user question while remaining robust to imperfect detections. To this end, we adopt a scoring-based frame selection strategy that leverages the LLM-predicted spatial cues, namely the key object set O_{key} and the cue object set O_{cue} . For each frame, we aggregate detection nodes’ confidence scores of objects from both sets using distinct weighting factors, reflecting their different semantic roles: key objects provide direct evidence of answer relevance, while cue objects contribute complementary contextual information. This weighted aggregation approximates query-conditioned semantic alignment under partial observability, enabling robust frame selection even when key objects are weakly detected or partially missing, and avoids the brittleness of single-object or maximum-confidence selection strategies. Finally, the user question and the k selected key frames across different locations and viewpoints are passed to the VLM for question answering.

4 Experiments

4.1 Implementation Details

We use YOLO-World [11] and BoT-SORT [2] for MemTree3D construction, following prior work [58] by adopting the ScanNet-200 [14] common indoor object categories. Additional implementation details are provided in the appendix. The tree construction runs online and in real-time (**25+ FPS**) on a single GPU, and can be easily integrated into embodied agents during the observation collection stage, in contrast to existing methods that rely on multiple vision foundation models [18, 29]. We set T_{thres} to 1.5 m and R_{thres} to 45° for all experiments, and fix the key-to-cue object weighting ratio to 10:1, without careful fine-tuning. While per-scene parameter tuning could further improve temporal segmentation and spatial retrieval, we adopt a unified setting across all scenes to better reflect real-world deployment, where the number of LocNode adapts automatically to scene scale. The LLM selects top k locations from MemTree3D after reasoning, which we default k to 3, with ablation study on different value of k in Fig 4.

4.2 Benchmark

We evaluate on OpenEQA [44], ScanQA [5] and SQA3D [42]. OpenEQA is a recent Embodied Question Answering (EQA) benchmark focusing on spatial

Table 1: Performance comparison of our MemTree3D with other SoTA methods on OpenEQA, including the performance on ScanNet and HM3D subset. Performance of MemTree3D with different number of used frame can be found in ablation study.

Method	Avg. Frame	LLM-Match		
		ScanNet	HM3D	ALL
<i>Open-source VLM</i>				
Video-LLaMA [60]	8	20.1	19.8	20.0
Video-ChatGPT [43]	100	32.9	30.4	32.1
LLaMA-2 w/ Concept Graph [44]	50	31.0	24.2	28.7
LLaMA-2 w/ Sparse Voxel Map [44]	50	36.0	30.9	34.3
LLaMA-2 w/ LLaVA-1.5 caption [44]	50	39.6	31.1	36.8
Qwen-2.5-VL-7B [6]	12	49.4	40.2	46.2
Video-LLaMA2 [12]	16	50.1	47.5	49.2
LLaVA-3D [63]	32	–	–	53.2
<i>Closed-source VLM</i>				
Claude-3 Opus [3]	20	–	–	36.3
Gemini 1.0 Pro Vision [49]	15	–	–	44.9
Claude-3.5 Sonnet [3]	20	–	–	48.7
GPT-4V w/ Concept Graph [44]	50	37.8	34.0	36.5
GPT-4V w/ Sparse Voxel Map [44]	50	40.9	35.0	38.9
GPT-4V w/ LLaVA-1.5 caption [44]	50	45.4	40.0	43.6
GPT-4V [1]	50	57.4	51.3	55.3
GPT-4o [28]	12	63.9	58.4	62.0
<i>Visual Search / 3D Memory Method</i>				
CLIP Retrieval [45]	3	55.1	41.9	50.6
3D-Mem [58]	3.1	–	–	57.2
VLM-Grounder [55]	6	65.1	58.8	63.0
<i>Ours (with zero-shot 2D VLM)</i>				
LLaVA-OneVision-7B [33]	3	52.5	42.9	49.2
MemTree3D (w/ LLaVA-OneVision-7B)	3	57.9 (+5.4)	49.3 (+6.4)	55.0 (+5.8)
GPT-4o [28]	3	55.8	37.1	49.4
MemTree3D (w/ GPT-4o)	3	69.4 (+13.6)	61.7 (+24.6)	66.8 (+17.4)

understanding and embodied reasoning. It contains 187 episode histories collected from ScanNet [14] and HM3D [46], with over 1,600 human-generated questions. Furthermore, OpenEQA adopts the automatic LLM evaluation protocol to evaluate the performance of the method. We follow the official setting and report the GPT-4 [1] LLM-Match score. ScanQA [5] and SQA3D [42] are another two large-scale benchmarks that focus on 3D scene spatial understanding, with ScanQA contains 4,675 and SQA3D contains 3,519 QA pairs. We follow previous works [23–25, 63] and evaluate on ScanQA validation and SQA3D test set using Exact Match (EM@1). More benchmark details and statistics can be found in our appendix.

4.3 Analysis on OpenEQA

In Table 1, we compared with multiple SOTA VLMs, captioning-based Socratic method [40], ConceptGraphs and Sparse Voxel Map [18] LLM methods on the OpenEQA benchmark. We also compared with three advanced visual search and 3D memory methods using GPT-4o as VLM, including detector-based VLM

Table 2: Category-level performance on OpenEQA.

Method	EQA Category						
	object recognition	object localization	attribute recognition	spatial understanding	object state recognition	functional reasoning	world knowledge
GPT-4o	54.0	36.9	58.3	36.8	54.6	53.4	50.2
w/ MemTree3D	64.6	58.1	68.9	52.5	75.4	73.6	72.3
Δ	+10.6	+21.2	+10.6	+15.7	+20.8	+20.2	+22.1

Table 3: EM@1 comparison on ScanQA and SQA3D.

Method	Avg. Frame	EM@1	
		ScanQA	SQA3D
<i>3D Fine-tuned Model</i>			
Scan2Cap [10]	–	–	41.0
ScanRefer+MCAN [9]	–	18.6	–
ClipBERT [32]	–	–	43.3
ScanQA [5]	–	21.1	47.2
3D-VisTA [64]	–	22.4	48.5
3D-LLM [20]	–	20.5	48.1
3D-VLP [30]	–	21.6	–
LEO [27]	–	24.5	50.0
ChatScene [23]	–	21.6	54.6
<i>VLM Method</i>			
Agent3D-zero [61]	24	17.5	–
LLaVA-Next-Video [39]	32	18.7	34.2
VideoChat2 [37]	16	19.2	37.3
LLaVA-OneVision-7B [33]	3	25.1	46.2
MemTree3D (w/ LLaVA-OneVision-7B)	3	28.0 (+2.9)	49.6 (+3.4)

Grounder [55], VLM-based frame selection 3D-Mem [58] and an image-text retrieval baseline with CLIP [45] that retrieve key frames by selecting the frame with highest image-query similarity.

In Table 1, we show our performance implemented with open-source VLM LLaVA-OneVision-7B and proprietary model GPT-4o on the OpenEQA benchmark. Our methods largely boost the performance of existing VLMs, with **5.8%** accuracy gain for LLaVA-OneVision, and **17.4%** over GPT-4o under the same number of input frame. Moreover, we achieve stronger performance compared with existing visual search methods using the same VLM GPT-4o.

In Table 2, we report our category-level performance on OpenEQA. Integrating MemTree3D consistently improves uniform sampling GPT-4o by more than 10% across all question categories. Although MemTree3D adopts an object-centric representation, its role is to retrieve informative key frames rather than perform fine-grained recognition. The final disambiguation of object attributes, states, and subtle visual details is handled by the VLM using raw visual evidence from the selected frames.

4.4 Analysis on ScanQA and SQA3D

On ScanQA and SQA3D (Table 3), we compared with 3D fine-tuned models, including task-specific models [5, 9, 10, 32] and 3D understanding LLMs [20, 23, 27,

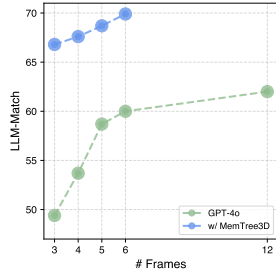


Fig. 4: OpenEQA results comparison with uniform sampling using different number of frames.

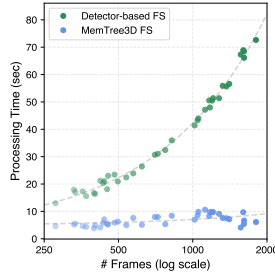


Fig. 5: Runtime comparison with Detector-based FS method. Measured after received question.

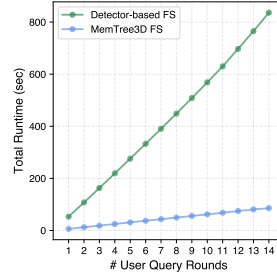


Fig. 6: Runtime comparison with Detector-based FS method when processing multi-round query.

Table 4: Ablation on performance using different LLM/VLM combinations.

LLM	VLM	ScanNet	HM3D	All
–	GPT-4o (Uniform Sampling)	55.8	37.1	49.4
Qwen3-4B	LLaVA-OneVision-7B	54.9	43.8	51.1
Qwen3-8B	LLaVA-OneVision-7B	56.8	46.0	53.1
GPT-4o	LLaVA-OneVision-7B	57.9	49.3	55.0
GPT-4o	GPT-4o	69.4	61.7	66.8

30, 64]. In Table 3, our method brings accuracy improvements and achieve strong zero-shot performance compared with existing 3D fine-tuned models and 2D VLM. However, we notice the performance gain on ScanQA and SQA3D is rather moderate compared with the large performance improvements on OpenEQA, which is likely caused by the smaller scene size in ScanNet, where the uniform sampling can already achieve decent visual coverage. This observation can also be justified by the smaller improvement on the ScanNet subset (avg. size $82.6 m^3$) compared to the HM3D subset (avg. size $556.0 m^3$) in OpenEQA, which suggests our method is more effective in a more challenging and larger 3D scene setting.

4.5 Ablation Studies

Does the number of frames affect performance? We evaluate the impact of using a larger k value in Fig 4. Our MemTree3D outperforms uniform sampling across different frame usage, highlighting the effectiveness of our method. Moreover, the performance of GPT-4o when using frames selected by MemTree3D consistently improves as more frames are included, due to access to richer visual information from diverse spatial locations in the scene.

How efficient is MemTree3D? To demonstrate that our proposed paradigm is more efficient than visual search key frame selection, we compare our proposed MemTree3D Frame Selection (MemTree3D FS) with the Detector-based

Table 5: Ablation on LocNode construction strategy.

Method	Avg. Frame Per Node	LLM-Match
Uniform	90	33.0
Uniform	30	40.4
6-DoF	36.3	49.1

Table 6: System robustness analysis under perception failure.

Query Object Presence	ScanNet	HM3D	All
✓	71.4	64.3	69.0
✗	67.9	62.1	65.9

Frame Selection (Detector-based FS) method [55] in Fig 5. We use the open source LLM Qwen3 [57] for both methods and run them on the same V100 GPU for fair comparison and reproducibility. The runtime of Detector-based FS increases with the number of video frames, while the runtime of MemTree3D FS remains relatively stable, with at least 69.2% runtime speed up compared to detector-based FS, highlighting the efficiency of our LLM key frame querying paradigm. The reported runtime is measured on OpenEQA, starting from the moment the system receives the user query. Furthermore, we compare the latency of MemTree3D and Detector-based FS in multi-round user query scenario in Fig 6, the latency gap highlights the better efficiency of our MemTree3D key frame selection paradigm compared with existing visual search approach. This efficiency stems directly from our design, which bypasses extensive per-query visual search and thereby minimizes computation.

Can MemTree3D work well with open source models? In Table 4, we conduct experiments and investigate the performance of various open source and proprietary LLM and VLM combinations with our MemTree3D. We found that switching proprietary LLM GPT-4o to open source smaller scale LLM Qwen3-4B and Qwen3-8B [57] and VLM LLaVA-OneVision-7B [33] does not lead to notable performance degradation, with LLM-Match score on OpenEQA outperforms uniform sampling proprietary VLM GPT-4o.

Ablation on Location Node construction strategy. We present an ablation study of different LocNode construction strategies in Table 5. The uniform construction strategy partitions Location Nodes using a fixed temporal window without considering camera motion, often grouping frames from distinct viewpoints into the same node. Increasing the node construction frequency by reducing the temporal window (from 90 to 30 frames) leads to improved performance, but still underperforms the 6-DoF-based construction, which explicitly leverages camera motion to segment the 3D scene into semantically coherent nodes, yielding a significant improvement of 8.7% over uniform sampling.

Is MemTree3D robust to perception failure? To evaluate the robustness of our framework under perception failures during MemTree3D construction, we analyze the performance gap with respect to **query object presence**, as reported in Table 6. We apply a heuristic string-matching strategy to determine whether object class names present in the constructed scene representation appear in the user question. On OpenEQA, our method achieves an LLM-Match score of 65.9 in the absence of query object presence, representing only a 3.1-point drop compared to cases where query objects are present.

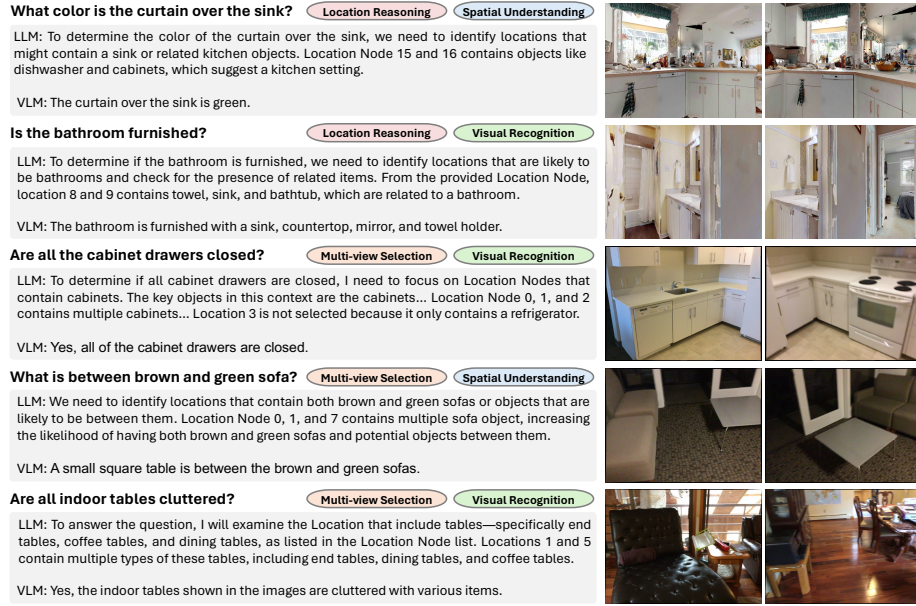


Fig. 7: Qualitative results. We present the top two selected key frames by our method, as well as the corresponding LLM location selection and VLM output.

4.6 Qualitative Results

We present qualitative results in Fig 7, which covers multiple question types such as location reasoning, multi-view selection, and spatial understanding.

Location Reasoning. When query-relevant key objects are missing from the MemTree3D representation due to perception failures, our pipeline uses the LLM to infer likely locations from the objects observed at each location.

Multi-view Selection. For questions requiring multiple viewpoints, our method retrieves frames from different location to provide comprehensive visual context.

Spatial Understanding and Visual Recognition. By selecting visually informative frames, our key-frame selection improves spatial reasoning and fine-grained visual recognition, resulting in more accurate answers.

5 Conclusion

We introduce the MemTree3D-guided key-frame selection paradigm for efficient and accurate 3D question answering. By constructing a compact 3D scene representation, MemTree3D enables LLM-based reasoning to select key frames without exhaustive visual search. Experiments on OpenEQA, ScanQA, and SQA3D show consistent improvements across both open-source and proprietary VLMs using fewer input frames, demonstrating the potential of our approach for real-world embodied AI applications.

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6 Appendix

We present more details and experiment results in the supplementary material structured as follows:

- Evaluation benchmark details.
- LLM prompt for key frame querying.
- More results on efficiency analysis.
- Additional qualitative results.
- Limitations and failure cases.

7 Evaluation Benchmarks Details

Table 7: Scale comparison of the evaluated 3D question answering benchmarks in our work.

Benchmark	# of scenes	# of questions
OpenEQA [44]	152	1,636
ScanQA [5]	<u>71</u>	4,306
SQA3D [42]	67	<u>3,519</u>

Table 8: Statistics of the ScanNet and HM3D subset from the OpenEQA. The table shows the number of scenes, questions, and the average scene size.

Subset	# scenes	avg. size (m^3)	avg. LocNode Δ	LLM-Match
ScanNet	89	82.6	5.9	+13.6
HM3D	63	556.0	20.3	+24.6

In this work, we evaluate on 3 different benchmarks, including OpenEQA [44], ScanQA [5], and SQA3D [42]. We provide detailed statistics of these benchmarks in Table 7. Among these benchmarks, OpenEQA has the most 3D scenes, featuring 3D scene collected from ScanNet [14] and HM3D [46]. And ScanQA having the largest scale with 4,306 questions.

We also compare the two subsets from OpenEQA in Table 8, including the number of 3D scene, average 3D scene size, average number of constructed LocNode, and the performance gain when using our MemTree3D. As discussed in our main paper, we notice a larger performance gain in the HM3D subset compare with the ScanNet subset, we believe this is caused by the larger 3D scene

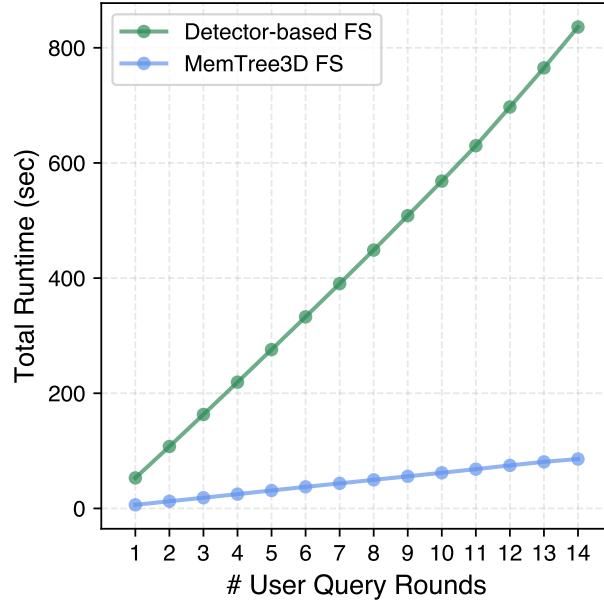


Fig. 8: Runtime comparison with Detector-based frame sampling method when processing multi-round query. Our method consistently achieve efficient key frame selection compared with detector-based frame selection approach.

size in HM3D. In ScanNet, the 3D scene size is much smaller, and uniform sampling serve as a strong baseline strategy to cover all the visual information from the 3D scene. However, in HM3D, the 3D scene size is much larger, and uniform sampling fails to cover all the visual details. Our proposed key frame selection strategy can address this challenges by selecting the most relevant key frame to the question, with a significant 24.6% performance gain on HM3D subset.

8 More detailed on efficiency analysis

We compare the runtime efficiency of our MemTree3D key frame selection framework against existing visual search-based approach, as shown in Fig 5. We also compare the efficiency analysis of multi-round query in Fig 8. Specifically, we measure the latency between receiving a user question and producing a response—an essential factor in real-world embodied AI scenarios. In such settings, an agent typically navigates in a 3D scene to collect visual observations and then answers user queries based on the visual stream. The key performance metric is the response time after receiving the user question. Visual search-based methods, such as VLM-Grounder [55], involve a multi-stage pipeline after receiving a question, resulting in substantial latency. The process typically includes:

1. Using an LLM to parse the question and extract a list of target objects relevant to the query.
2. Running open-vocabulary object detector on all video frames using the identified target objects as text prompts.
3. Sending the subset of frames containing detected objects to a VLM for question answering.

Among these steps, step 2 leads to large latency issue due to the extensive computational cost growing proportionally with the number of video frames. In contrast, our tree based frame selection framework pre-builds the 3D representation before receiving the user’s question, instead of searching the answer in the video frames, our work searches the key frame from the constructed MemTree3D, therefore bypasses this extensive visual search process, and incurs minimal latency regardless of the video length, as illustrated in Fig 5.

9 Implementation Details

LLM/VLM Inference Parameters. We set the inference temperature of GPT-4o and LLaVA-OneVision-7B to 0.0, and 0.6 for Qwen3 following the default setting. The rest of the inference parameters follow the default generation configuration from their corresponding huggingface repositories and official API setting.

LLM Location Selection Prompt. We present the full prompt used for location selection with GPT-4o in Fig 9. The prompt includes a one-shot example illustrating the expected JSON response format, along with custom tags such as <think> and <answer> to guide the model’s reasoning and output parsing.

Image-text Retrieval Baseline. We implement the Image-text retrieval baseline using CLIP [45] from the openai-clip-vit-base-patch32 checkpoint. We directly use the user question as input text and compare the image-text cosine similarity to retrieve the top k images for GPT-4o VLM input.

10 Additional Qualitative Results

We present more qualitative results in Fig 10. With more frame selection results comparison with VLM image-text retrieval baseline method [45]. Here, we present several failure cases of the VLM image-text retrieval baseline, including 1. VLM fails to retrieve any key frames related to question (row 1), where the key frame selection required question reasoning and location understanding. 2. VLM samples key frames from the similar location and viewpoint, and failed to retrieve correct key frame for the question (row 2-4), as direct selecting top-k frames based on similarity score can lead to over sample on the similar location and viewpoint, our MemTree3D segments the 3D scene into multiple LocNode and alleviates this issue.

11 Limitations and Failure Cases

Despite the better efficiency and significant performance gain over existing visual search key frame selection method, our MemTree3D still has its limitation. Our tree based key frame selection paradigm can occasionally fail in some challenging object localization question where the object is not directly detected. More specifically, when the object localization question is querying an entirely novel object that is not part of MemTree3D, the LLM during key frame querying process will make its most educated guess to select several most possible locations that the target object might appear, but this best effort of reasoning does not always yield successful object localization. We illustrate several failure cases in Fig 11, where the LLM selects several reasonable locations but fails to retrieve the target objects.

Location Node (LocNode): contains Object Node from multiple locations with similar spatial coordinates. It contains a unique Location ID.

Object Node (ObjNode): contains an object's multiple Detection Node across a period of time span. Object Node include information such as object class name, object tracking ID.

Now, I will prompt you a question, and you will be given a list of Location Node.

Based on the provided Location Node list, observe the object node contain in each location node, select <top_k_locs> location node that you think might contain critical visual information that can help answer the given question.

Next, select Object Node within each Location Node that belongs to Key objects (critical object that is related to the answer)

And then, select Object Node within each Location Node that belongs to Cue objects (objects that might be near key objects)

Provide the Location ID (key) and key_objects and cue_objects (value) in the following format.

<think> Conclude the <top_k_locs> location you want to select based on your best guess. You must select <top_k_locs>!</think>

<answer>

```
{
  "0": {
    "key_objects": ["tv"],
    "cue_objects": ["picture", "clock"]
  },
  "1": {
    "key_objects": ["tv"],
    "cue_objects": ["picture", "desk"]
  },
  "2": {
    "key_objects": ["tv", "monitor"],
    "cue_objects": ["picture", "tv stand"]
  }
}
```

</answer>

Question: <question>

Location Node list : <MemTree3D_JSON>

Fig. 9: LLM prompt for MemTree3D querying and location selection.

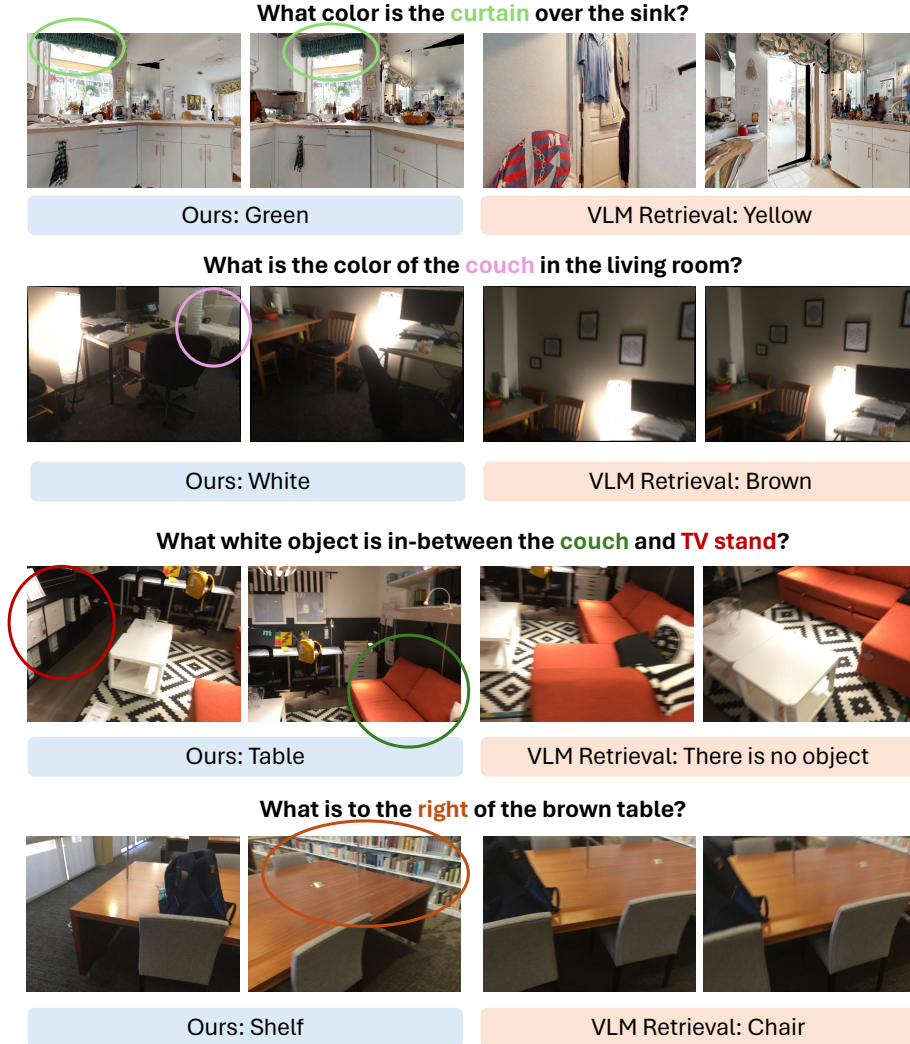


Fig. 10: Additional qualitative results comparison between our MemTree3D framework and VLM retrieval baseline. We showcase several cases, including 1. VLM fails to retrieve any key frames related to question (row 1). 2. VLM samples key frames from the similar location and viewpoint, and failed to retrieve correct key frame for the question (row 2-4).

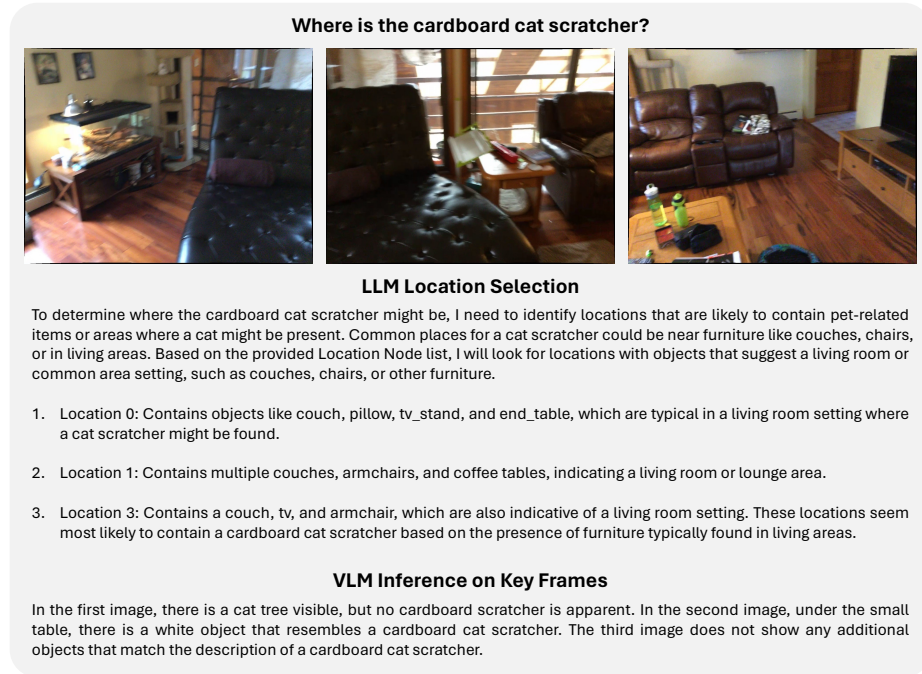


Fig. 11: A failure case of an novel object localization question from the OpenEQA. The question asked to locate where is the cardboard cat scratcher, because the cat scratcher is not part of the `ObjNode` in `MemTree3D`, our LLM conduct its best reasoning effort to select the most possible locations that the cat scratcher can be. However, the final selected frames does not contain a clear view for VLM to answer the question.